TASK 3: ABS WHEEL SPEED SENSOR DIAGNOSTIC INVESTIGATION

Autocare Technician Level 2 | Block 8 | v2.1

TIME	EVIDENCE	PROGRAMME POINT
60 minutes	**Practical | Knowledge | Documentation**	**Weeks 71-80**

Your task

A vehicle has an ABS or traction warning lamp. You must carry out a structured inspection and test of the wheel speed sensor circuit or output, record evidence and explain the likely fault or referral route.

What you must do

1. Prepare the vehicle/rig and confirm the warning lamp or fault condition.
2. Read and record relevant DTCs if equipment is available.
3. Inspect the wheel speed sensor, wiring and reluctor/tone ring as instructed.
4. Check output using a suitable method and record results.
5. Explain findings and reinstate the vehicle/workstation.

What good evidence looks like

SAFE SETUP

Suitable PPE, a prepared work area and controlled vehicle or equipment.

CORRECT METHOD

A logical sequence using the relevant procedure and technical information.

ACCURATE EVIDENCE

Actual readings, findings and actions recorded clearly and precisely.

SOUND DECISIONS

Appropriate recommendations, referral or authorisation supported by evidence.

You will need

- Light vehicle or ABS training rig
- Scan tool, multimeter or oscilloscope if available
- Manufacturer technical information
- PPE and Apprentice Assessment Record

SOW ALIGNMENT

Block 8 - Advanced Diagnostics and Workshop Management
Duties: 1-10 KSB evidence focus: K15-K21, S10-S15, B1, B2
Task reference: Block_8_Task_3_ABS_Wheel_Speed_Diagnostic

TASK 3: ABS WHEEL SPEED SENSOR DIAGNOSTIC INVESTIGATION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

TASK COMPLETED AT

○ **APPRENTICE'S WORKPLACE**

○ **TRAINING CENTRE**

Vehicle make

Model

Registration

Mileage

Engine size / type

VIN / chassis number

Measurements and technical checks

DTC code		Sensor type	
Expected waveform/output		Measured output	
Air gap if relevant		Wiring condition	
Tone ring condition		Related systems affected	

Special tools, equipment or calibration requirements

Task evidence

PPE and precautions used

Safety-critical findings and immediate action

Work completed, findings and recommendations

Knowledge check

1. What is the purpose of an ABS wheel speed sensor?

2. What does the term air gap mean?

Knowledge check - continued

3. (a) Name causes of an illuminated ABS warning lamp. (b) Why can ABS faults affect other vehicle systems?

4. Why may an oscilloscope provide better information than a basic meter?

5. (a) How would you explain an ABS fault to a customer? (b) What further tests or authorisation may be required?

Assessor/Workplace Mentor observation

- ☐ Carry out safe ABS diagnostic checks.
- ☐ Identify ABS sensor type and relevant components.
- ☐ Record DTCs, signal outputs or visual findings.
- ☐ Explain how the evidence supports a diagnosis.
- ☐ Identify effects on related systems and customer communication.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

○ **PASS** ○ **REFER**

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

PIN (if applicable)

Apprentice signature

Assessor/Workplace Mentor signature

Date